HH-I-545B
March 4, 1971
SUPERSEDING
Fed. Spec. HH-I-00545A(GSA-FSS)
August 8, 1968

FEDERAL SPECIFICATION

INSULATION, THERMAL AND ACOUSTICAL (MINERAL FIBER, DUCT LINING MATERIAL)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers mineral fiber insulation for lining the interior surfaces of ducts, plenums, and other air-handling equipment, and to provide sound attenuation in systems that handle air up to 250° F.

1.2 Classification. The insulation shall be of the following types, as specified (see 6.2).

Type 1 - Flexible, flat.
Type II - Nonflexible, flat.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal from a part of this specification to the extent specified herein.

Federal Specifications:

UU-P-268 - Paper, Kraft, Untreated, Wrapping.
PPP-B-601 - Box, Wood, Cleated Plywood.
PPP-B-636 - Box, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Tripple Wall.
PPP-B-1055 - Barrier, Material, Waterproofed, Flexible.
PPP-T-76 - Tape, Pressure Sensitive Adhesion Paper (For Carton Sealing).

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Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of the specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

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American Society for Testing and Materials (ASTM) Standards:

C167 - Thickness and Density of Blanket or Bolt-Type Thermal Insulating Materials.
C177 - Thermal Conductivity of Materials by means of the Guarded Hot Plate.
C423 - Sound Absorption of Acoustical Materials in Reverberation Room.
C553 - Mineral Fiber Blanket and Felt Insulation (Industrial Type).
E84 - Surface Burning Characteristics of Building Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103).

Department of Commerce

CS 131 - Industrial Mineral Wool Products, All Types, Testing and Reporting.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402).

National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., Agent, 1616 P Street N.W. Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 202 Union Station, 516 W. Jackson Blvd., Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. The basic material shall be mineral fibers of rock,

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slag, or glass composition, or mixtures thereof processed from a molten state into fibrous form with a binder added to form dimensionally stable, flexible, or nonflexible insulation.

3.2 Dimension and form. The insulation shall be of the following sizes as specified (see 6.2): Type I, flexible, flat in rolls up to 200 feet in length, 36 to 72 inches in width, and thicknesses of 1/2 inch to 3 inches, in 1/2 inch increments. Type II, nonflexible, flat, in sheet form, up to 60 inches in length, up to 48 inches in width, and in thickness of 1/2 to 3 inches in 1/2 inch increments.

3.3 Physical characteristics.

3.3.1 Flexibility, type I. When tested in accordance with 4.4.1, insulation that shows no visible rupture on the outer surface, shall be considered type I, flexible. Insulation that shows visible rupture shall be considered type II, nonflexible.

3.3.2 Density. When tested in accordance with 4.4.3, the average density of type I insulation, including surface coating or facing, shall be not less than 1.35 pounds per cubic foot nor greater than 3.3 pounds per cubic foot. The average density of type II shall be not less than 2.7 pounds per cubic foot nor greater than 8.8 pounds per cubic foot.

3.3.3 Corrosiveness. When tested in accordance with 4.4.4, the steel test plate in contact with the insulation shall show no more corrosion than sterile cotton in contact with a steel plate.

3.3.4 Moisture absorption. When tested in accordance with 4.4.5, the insulation shall gain not more than 0.5 percent by volume.

3.3.5 Mold growth. When tested in accordance with 4.4.6, the insulation shall be resistant to the effects of high humidity under ordinary atmospheric temperature conditions. The mold shall not spread beyond the inoculated area and no significant growth shall be observed.

3.3.6 Temperature resistance. When tested in accordance with 4.4.7, the insulation, including adhesives shall be resistant to the effects of temperature at 250° F. and shall not deform or delaminate.

3.3.7 Erosion resistance. When tested in accordance with 4.4.8, the insulation shall not break away, flake off, or show evidence of delamination or continued erosion when air is passed through typical sections at a velocity of 2-1/2 times the manufacturer's rated velocity.

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3.3.8 Odor emission. When tested in accordance with 4.4.9, the insulation shall not emit odor unacceptable to the test panel.

3.3.9 Surface burning characteristics. When tested in accordance with 4.4.10, the insulation, with the side intended for exposure to moving air exposed to flame, shall have a maximum flame spread rating of 25 and a maximum smoke developed rating of 50. If the insulation is to be applied with adhesives, it shall be tested as applied with such adhesives, or the adhesives shall have a flame-spread rating not over 25, and a smoke-developed rating not higher than 50.

3.3.10 Thermal conductance. When tested in accordance with 4.4.11, the thermal conductance (C) of the insulation, expressed ad B.T.u. hr. sq. ft. deg. F. for the specified thickness shall not exceed the values shown in table I.

TABLE I. Thermal conductivity, maximum

Mean Temperature (°F.)	Type I		Type II	
	1/2 inch	1 inch	5/8 inch	1 inch
50	0.54	0.27	0.40	0.25
75	.58	.29	.43	.27
100	.64	.32	.47	.29

3.3.11 Sound absorption coefficients. When tested in accordance with 4.4.12, with the test sample mounted on 24-gauge sheet steel, supported by 1 inch by 1 inch by 1/8 inch steel angle, the insulation up to and including 2 inches shall have sound absorption coefficients at specified frequencies not less than those in table II (see 6.2).

TABLE II. Sound absorption coefficients

	Thickness	Frequency, Hertz					
	Inch	125	250	500	1000	2000	4000
Type I	1/2	0.15	0.40	0.34	0.47	0.60	0.74
	1	.25	.53	.54	.70	.80	.80
	1 1/2	.20	.61	.73	.89	.80	.80
	2	.25	.68	.86	.90	.80	.85
Type II	5/8	.12	.40	.34	.65	.76	.75
	1	.13	.44	.63	.90	.83	.76
	1 1/2	.15	.60	.80	.92	.83	.82
	2	.30	.63	.94	.92	.81	.80

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3.3.12 Tolerances. The tolerances for flat insulation shall be minus 1/8 inch for length, and width. The tolerances for insulation of 5/8 inch thickness or less shall be minus 0; for insulation 1 inch or greater shall be minus 1/8 inch.

3.4 Workmanship. The insulation shall have true dimensions, square corners, parallel sides and shall be free of grease, oil, dirt, and other foreign matter. The insulation shall also be free of visible defects which might interfere with use of the material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. A lot shall consist of each type and size of insulation offered for delivery at one time.

4.2.2 Visual and dimensional examination. Sampling for this examination shall be in accordance with level S-3 in MIL-STD-105. The sample unit shall be one completely fabricated roll or sheet.

4.2.3 Sampling for tests.

4.2.3.1 Sampling for density. The number of samples for the density test shall be in accordance with level S-2 in MIL-STD-105. The sample unit shall be one completely fabricated roll or sheet.

4.2.3.2 Samples for test. Samples for all tests except density shall be in accordance with level S-1 of MIL-STD-105. The sample unit shall be one completely fabricated roll or sheet.

4.3 Visual and dimensional examination. Insulation shall be examined for defects listed in table III. The acceptable quality level shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

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TABLE III. Classification of defects

Defects	Major	Minor
Type not as specified	X	
Corners and edges not square	X	
Sides not parallel	X	
Hole in insulation or ragged edges	X	
Dimensions not as specified or not within tolerances	X	
Not free from grease, oil, dirt or other foreign matter		X
Damage or defects affecting function or serviceability	X	
Damage or defects but not affecting function or serviceability		X

4.4 Tests. Tests for temperature, erosion resistance, surface burning characteristics, and sound absorption shall be conducted on the side of material intended for exposure to moving air.

4.4.1 Flexibility and rigidity.

4.4.1.1 Flexibility, type I. A 12-inch by 12-inch by 1 inch piece of insulation shall be bent over a 1/2-inch steel pipe through an angle of 90 degrees, then released.

4.4.1.2 Rigidity, type II. A piece of insulation 32 inches long, 1 inch thick, and between 6 and 24 inches wide, shall be placed on two horizontal, parallel, 1/2-inch pipes spaced 30 inches apart. The sag at the center of the span shall be measured from a straight line connecting two points on the surface directly above the supports.

4.4.2 Dimensions. Dimensions shall be determined in accordance with ASTM C167.

4.4.3 Density. Determination of density at specified thickness shall be in accordance with ASTM C167.

4.4.4 Corrosiveness. Corrosiveness shall be determined in accordance with section III of CS 131.

4.4.5 Moisture adsorption. Determination of moisture adsorption shall be in accordance with ASTM C553.

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4.4.6 Mold growth. Three 4- by 4-inch samples shall be inoculated with Mold Mycelia and Spores from bread, and then placed in a closed container at 100% relative humidity and room temperature for a period of 60 days. The samples shall be examined for the extent of mold growth. The mold shall not spread beyond the inoculated area and no significant growth shall be observed. As a referee method, Chaetomium Globasium may be used instead of Mold Mycelia.

4.4.7 Temperature resistance. Three 8- by 8- by 8-inch samples shall be placed in a closed vessel at 100% relative humidity and room temperature for a period of 24 hours. The samples shall then be tested by simultaneous exposure to air maintained at not less than 125° F. on the exterior surface and not less than 265° F. on the interior surface. Any practical arrangement may be employed. The samples shall then be examined visually for indications of deterioration.

4.4.8 Erosion resistance. A sample of one thick insulation is installed in a 12- by 12-inch sheet metal duct in an L-shaped assembly. The assembly is connected to a fan by means of a transformation piece to provide for uniform air entry to the test sample. The test velocity shall be 2-1/2 times the maximum velocity recommended by the manufacturer but not less than 10,000 feet per minute of air at room temperature, measured by a pilot tube. Room air, between 60° and 100° F. may be employed. The inlet of the air shall be covered with a double layer of cheesecloth known to the trade as "count of 32 by 24." A damper, if employed should be located between the screen and the fan or between the fan and the inlet to the test sample. A collecting screen sized to provide for an area not less than 5 times the inside cross-sectional area of the test sample is to be located 12 inches from the outlet end of the test apparatus. The relative humidity should be less than 70% for 24 hours prior to the test. Air is passed through the assembly at test velocity with the collecting screen removed for a period of one hour. The screen is placed in position and the test is continued for four hours. The screen is to be examined for macroscopic particles at the end of each hour. At the end of the test period evidence of continued erosion is not to be evident, interior surfaces of the sample are not to show evidence of cracking, flaking, peeling, or delamination.

4.4.9 Odor emission. A minimum of one square foot sample of insulation shall be subjected to a soaking temperature of 150° F. for 15 minutes in a closed two gallon metal container and examined

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by a penal of 5 persons normal odor perception. The responses of this panel to the following questions shall be noted:

(1) Was the odor perceptible?

(2) Was the odor objectionable?

(3) Was the odor weak, strong, or very strong?

A detactable odor of objectionable nature recorded by more than 1 of the 5 panel members would constitute grounds for nonacceptance of the insulation.

4.4.10 Surface burning characteristics. Determination of surface burning characteristics shall be in accordance with ASTM E84.

4.4.11 Thermal conductance. Determination of thermal conductance shall be in accordance with ASTM C177.

4.4.12 Sound absorption coefficients. Determination of sound absorption coefficients shall be in accordance with ASTM C423.

4.5 Inspection of preparation for delivery. The preservation, packaging, packing, and marking of the unit shall be inspected to determine conformance with the applicable requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Type I. Type I insulation having similar dimensions and composition shall be furnished in rolls and shall be wrapped with barrier material conforming to PPP-B-1055, class E-2, and waterproofed-sealed with tape conforming to PPP-T-76.

5.1.1.2 Type II. Type II insulation having similar dimensions and composition shall be packaged in a fiberboard box conforming to PPP-B-636, type CF, class weather-resistant, grade W5c, except that box dimension limitations are waived for larger sizes.

5.1.2 Level B (see 6.4).

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5.1.2.1 Type I. Type I insulation of like description furnished in rolls shall be wrapped with paper conforming to UU-P-268, grade B and secured with tape.

5.1.2.2 Type II. Type II insulation of like description shall be packaged in a close-fitting box conforming to PPP-B-636, type CF, class domestic. The box shall be closed in accordance with the appendix to the box specification.

5.1.3 Level C. The insulation shall be packaged in accordance with the supplier's commercial practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Insulation packaged as specified in 5.1 shall be packed in plywood boxes conforming to PPP-B-601, overseas type.

5.2.2 Level B.

5.2.2.1 Military requirements. Insulation packaged as specified in 5.1 shall be packed in boxes conforming to PPP-B-640, weather-resistant or PPP-B-636, V3c, except that box dimensions limitations are waived for larger sizes.

5.2.2.2 Civil agency requirements. Type I insulation of like description, packaged as specified in 5.1.2.1, shall be packed in a close-fitting box conforming to PPP-B-636, class domestic or PPP-B-640, Type II insulation there is no packing required.

5.2.3 Level C. The insulation shall be packed to insure carrier acceptance and safe delivery to destination at lowest rates in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages, and shipping containers shall be marked in accordance with Fed.Std. 123 or MIL-STD-129 as applicable (see 6.2).

6. NOTES

6.1 Intended use. All types of insulation are intended for use on interior surfaces of ducts, plenums, and other air-handling equipment to allow minimum air friction and surface erosion and to effectively absorb sound energy within the audible frequency range.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number and date of this specification.

b. Type of insulation (see 1.2).

c. Length, width, thickness required (see 3.2).

d. Sound absorption coefficient frequency (see 3.3.11).

e. Level of packaging and level of packing required (see 5.1 and 5.2).

f. Marking required (see 5.3).

6.3 Air surface of insulation. Depending on the insulation characteristics and minimum service air velocity, the air surface of insulation may be additionally coated with suitable binder material, faced with fibrous mat, or faced with fabric and coated.

6.4 Level B packaging (see 5.1.2) hss been developed by GSA for civil agency procurement. For Navy procurement details of level B packaging, when required, should be developed at the time of procurement when all conditions are known.

MILITARY CUSTODIANS:

Army - ME
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Review activities:
Army - ME, CE
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CIVIL AGENCIES INTEREST:

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

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c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* -- Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* -- Optional
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